

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/ Marks
CO)

Q. 1 Attempt the following questions.

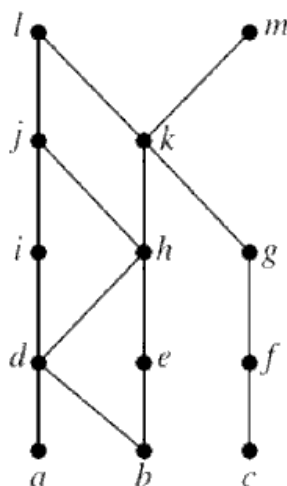
[12]

- A) Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a professor," "x is ignorant," and "x is vain," respectively. Express each of these statements using quantifiers; logical connectives; and $P(x)$, $Q(x)$, and $R(x)$, where the domain consists of all people. **CO1**
- (a) No professors are ignorant. (b) All ignorant people are vain.
(c) No professors are vain. (d) Does (c) follows (a) and (b)?
- B) Show that if n is a positive integer, then $1+2+3+\dots+n=\frac{n(n+1)}{2}$ by using mathematical induction. **CO1**
- C) Use De Morgan's laws to find the negation of each of the following statements. **CO1**
- (a) Kwame will take a job in industry or go to graduate school.
(b) Yoshiko knows Java and calculus.
(c) James is young and strong.
(d) Rita will move to Oregon or Washington.

Q.2 Solve Any Two of the following.

[12]

- A) Answer these questions for the partial order represented by this Hasse diagram. **CO2**



- a) Find the maximal elements.
- b) Find the minimal elements.
- c) Is there a greatest element?
- d) Is there a least element?
- e) Find all upper bounds of $\{a, b, c\}$.
- f) Find all lower bounds of $\{f, g, h\}$.

B) Let $f(x) = x + 2$, $g(x) = x - 2$, and $h(x) = 3x$ for $x \in \mathbb{R}$ where $\mathbb{R}$ = set of real numbers. Find **C02**

i) gof

ii) fohog

iii) foh

C) Solve the following homogeneous recurrence relation: **C03**

$$t_n = 4t_{n-1} - 4t_{n-2}.$$

$$t_n = 1, n = 0 \text{ and } n = 1.$$

Q.3 Solve Any Two of the following.

[12]

A) Draw the following graphs. **C04**

a) K_7

b) $K_{1,8}$

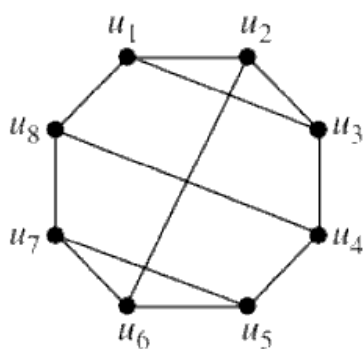
c) $K_{4,4}$

d) C_7

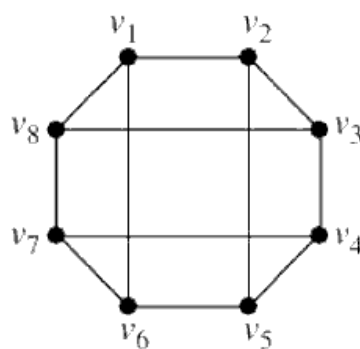
e) W_7

f) Q_4

B) Use paths either to show that these graphs are not isomorphic or to find an isomorphism between them. **C04**

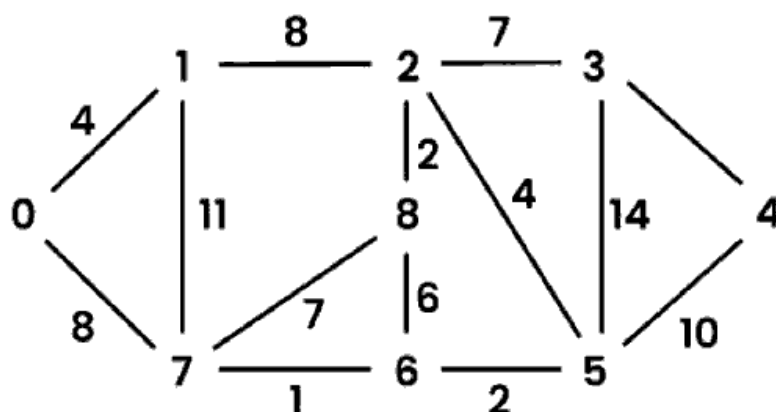


G



H

C) Use Dijkstra's algorithm to find the length of a shortest path between the vertices 1 and 4 in the weighted graph displayed in Figure below. **C04**



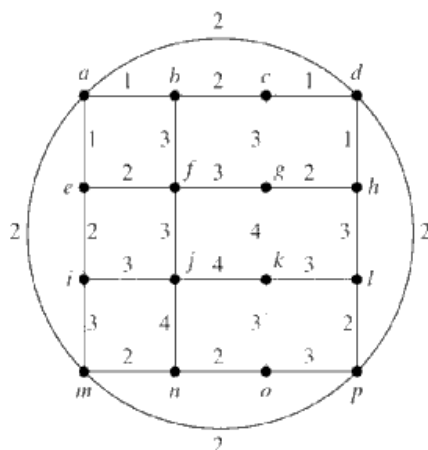
Q.4 Attempt the following questions.

[12]

A) Use Huffman's algorithm to provide an optimal average-bit-length code for the probability distribution a: 0.2, b: 0.2, c: 0.15, d: 0.15, e: 0.15, f: 0.1, and g: 0.05. Draw a binary tree to find the prefix code for each probability distribution **C04**

and compute the average bit-length of a codeword.

- B)** Use Kruskal's algorithm to find a minimum spanning tree for the weighted graph in **C04** the Figure below.



- C)** (i) What is the value of each of the prefix expressions $+ - \uparrow 3 2 \uparrow 2 3 / 6 - 4 2$? **C04**
(ii) What is the value of each of the postfix expressions $3 2 * 2 \uparrow 5 3 - 8 4 / * -$?

Q. 5 Attempt the following questions.

[12]

- A)** Consider the binary operation defined on the set $A = \{a, b, c, d\}$ by following table. **C05**
Find:

*	a	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

- (i) $c * d$ and $d * c$
(ii) $b * d$ and $d * b$
(iii) $a * (b * c)$ and $(a * b) * c$
- B)** Consider the group $G = \{1, 2, 4, 7, 8, 11, 13, 14\}$ under multiplication modulo 15. **C05**
i) Find multiplication table of G .
ii) Find 2^{-1} , 7^{-1} .
iii) Find the orders and subgroups generated by 7 and 11.

*** End ***